

3. X-rays are commonly used to diagnose illness.

(a) Refer to **Figure 2** to answer the following.

It is stated that X-rays have shorter wavelengths than ultraviolet rays but longer wavelengths than gamma rays. Explain whether you agree using data to support your answer. [3]

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(b) Refer to **Figure 3** to answer the following.

A patient has a series of X-rays taken following an accident. These were: one chest; one abdomen; and one skull X-ray. Calculate the percentage of the yearly background radiation the patient received from this combination of X-rays. [3]  
(1 year = 365 days)

percentage = .....

